Chapter 9. Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
- Partitioning Methods
- Hierarchical Methods
- Density-Based Methods
- Expectation Maximization
- Evaluation of Clustering 
- Summary

Determine the Number of Clusters

- Empirical method
  - # of clusters ≈√n/2 for a dataset of n points
- Elbow method
  - Use the turning point in the curve of sum of within cluster variance w.r.t the # of clusters
- Cross validation method
  - Divide a given data set into *k* parts
  - Use *k* – 1 parts to obtain a clustering model
  - Use the remaining part to test the quality of the clustering
    - E.g., For each point in the test set, find the closest centroid.
    - Consequently, one can use the sum of squared distances between all points in the test set and the closest centroid**s** to measure how well the model fits the test set.
  - Compare the overall quality measure w.r.t. different *k's*, and **find number of clusters** that fits the data the best.

Measuring Clustering Quality

- Two methods: extrinsic vs. intrinsic
- Extrinsic: supervised, i.e., the ground truth is available
  - Compare a clustering against the ground truth using certain clustering quality measure
  - Ex. Bcubed precision and recall metrics
- Intrinsic: unsupervised, i.e., the ground truth is unavailable
  - Evaluate the goodness of a clustering by considering how well the clusters are separated, and how compact the clusters are.
  - Ex. Silhouette coefficient

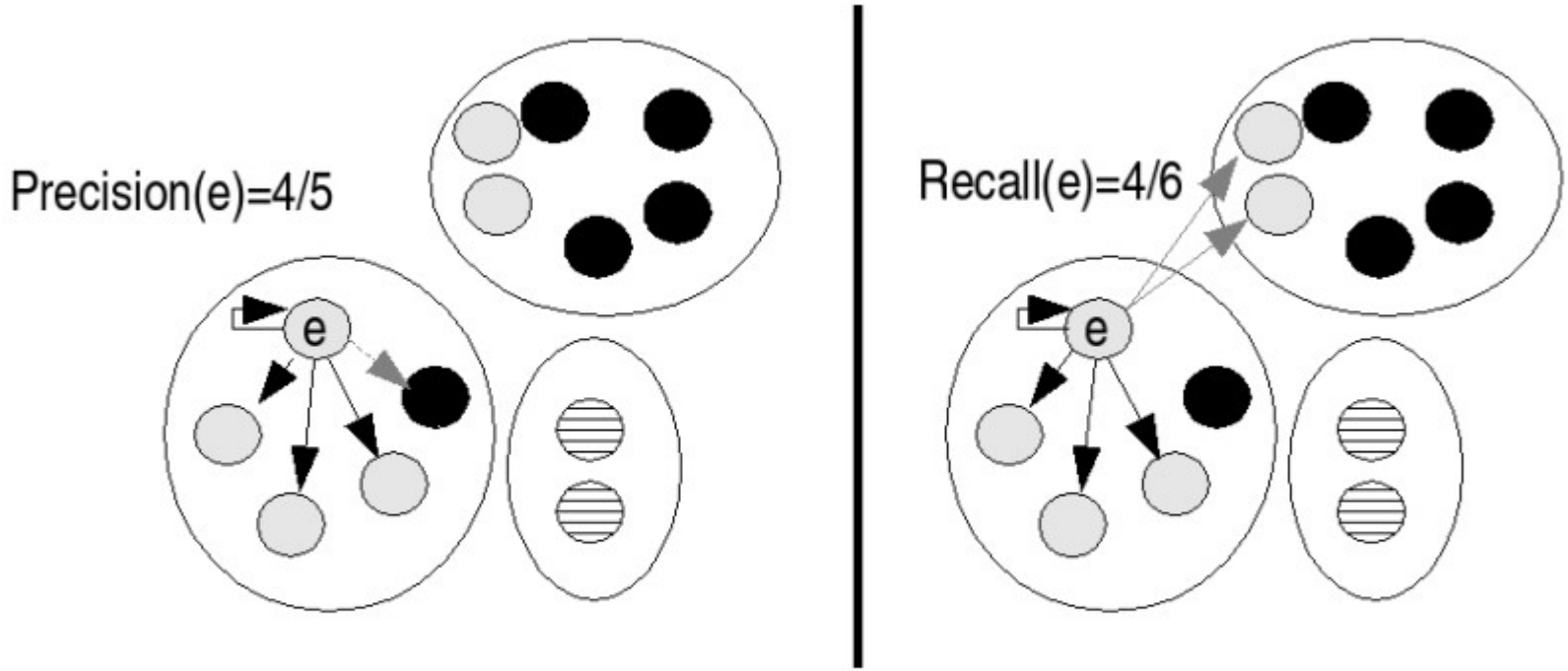
Bcubed precision and recall metrics

- BCubed evaluates the precision and recall for every object in a clustering on a given data set according to ground truth.

- The precision of an object indicates how many other objects in the same cluster belong to the same category as the object.

- The recall of an object reflects how many objects of the same category are assigned to the same cluster

Bcubed precision and recall - example



Example of computing the BCubed precision and recall for one item

Silhouette coefficient

- The silhouette coefficient is such a measure. For a data set, D, of n objects, suppose D is partitioned into k clusters, C_1, ... , C_k. For each object o ∈ D, we calculate a(o) as the average distance between o and all other objects in the cluster to which o belongs. Similarly, b(o) is the minimum average distance from o to all clusters to which o does not belong. Formally, suppose o ∈ C_i (1 ≤ i ≤ k); then

$$a(o) = \frac{\sum_{o' \in C_i, o \neq o'} dist(o, o')}{|C_i| - 1}$$

$$b(o) = \min_{C_j : 1 \leq j \leq k, j \neq i} \left\{ \frac{\sum_{o' \in C_j} dist(o, o')}{|C_j|} \right\}$$

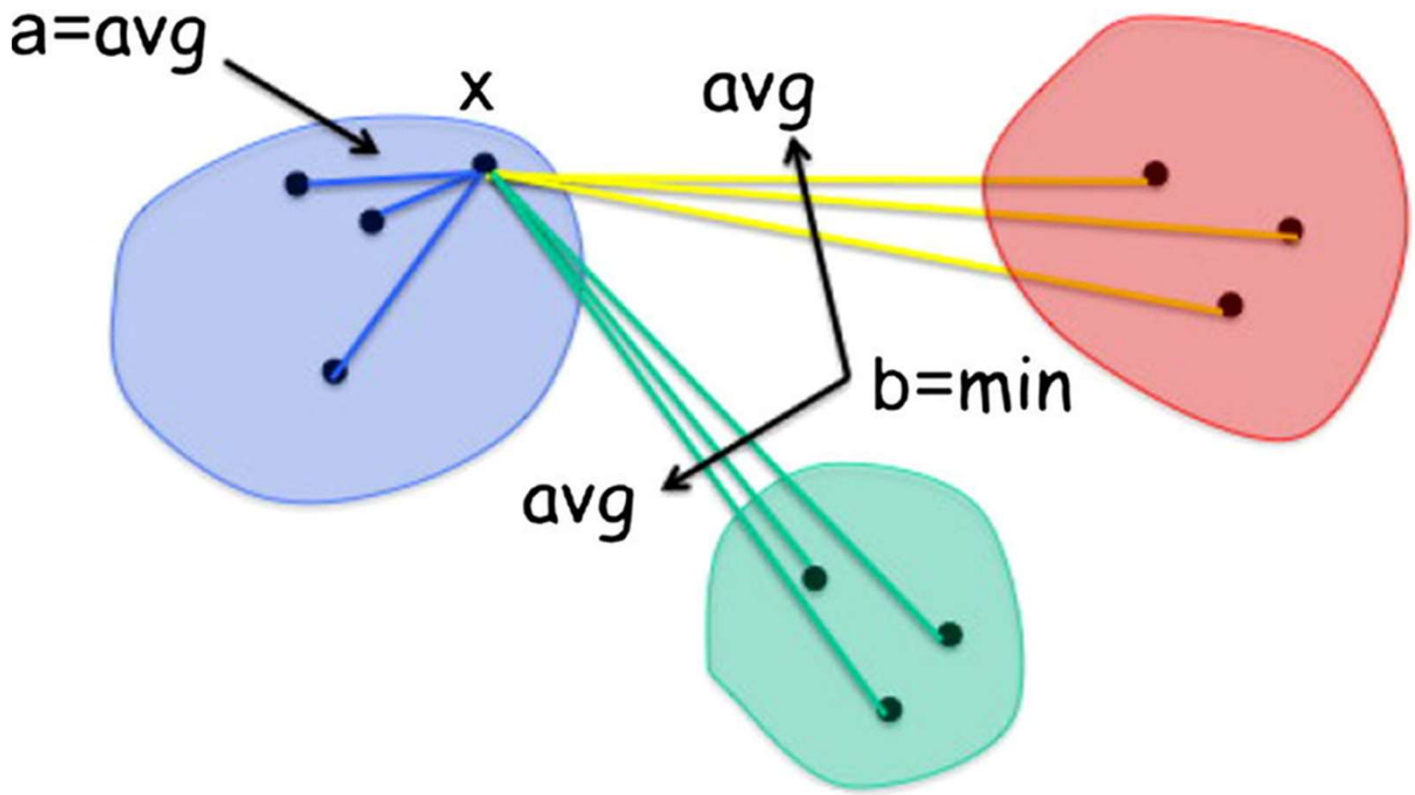
- The silhouette coefficient of o is then defined as

$$s(o) = \frac{b(o) - a(o)}{\max\{a(o), b(o)\}}$$

Silhouette coefficient (cont.)

- The value of the silhouette coefficient is between −1 and 1.
- The value of a(o) reflects the compactness of the cluster to which o belongs. The smaller the value, the more compact the cluster.
- The value of b(o) captures the degree to which o is separated from other clusters. The larger b(o) is, the more separated o is from other clusters.
- Therefore, when the silhouette coefficient value of o approaches 1, the cluster containing o is compact and o is far away from other clusters. However, when the silhouette coefficient value is negative (i.e., b(o) < a(o)), this means that, o is closer to the objects in another cluster than to the objects in the same cluster as o.
- To measure a cluster's fitness within a clustering, one can compute the average silhouette coefficient value of all objects in the cluster. To measure the quality of a clustering, one can use the average silhouette coefficient value of all objects in the data set.

Silhouette coefficient - example



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Summary

- Cluster analysis groups objects based on their similarity and has wide applications
- Measure of similarity can be computed for various types of data
- Clustering algorithms can be categorized into partitioning methods, hierarchical methods, density-based methods, grid-based methods, and model-based methods
- K-means and K-medoids algorithms are popular partitioning-based clustering algorithms
  - Birch and Chameleon are interesting hierarchical clustering algorithms, and there are also probabilistic hierarchical clustering algorithms
- DBSCAN, OPTICS, and DENCLUE are interesting density-based algorithms

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